
NYCU IMF - 114 Financial Econometrics

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Assignment: 3, Ch4 q1, 3, 25

Question 1

Answer each of the following:

- (a) Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2 .

Recapitulate:

- SST : Total Sum of Squares = $\sum (y_i - \bar{y})^2$
- SSR : Sum of Squares Regression (Explained Sum of Squares) = $\sum (\hat{y}_i - \bar{y})^2$
- SSE : Sum of Squared Errors (Residual Sum of Squares) = $\sum \hat{e}_i^2$
- $SST = SSR + SSE$
- $R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$

Given: $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$.

Step 1: Calculate Total Sum of Squares (SST)

$$SST = \sum (y_i - \bar{y})^2$$
$$SST = \sum y_i^2 - N\bar{y}^2$$

Substitute the given values into the formula:

$$SST = 7825.94 - 20(19.21)^2$$
$$SST = 7825.94 - 20(369.0241)$$
$$SST = 7825.94 - 7380.482$$
$$SST = 445.458$$

Step 2: Calculate R^2 Using the formula for R^2 based on the Regression Sum of Squares (SSR):

$$R^2 = \frac{SSR}{SST}$$
$$R^2 = \frac{375.47}{445.458}$$
$$R^2 \approx 0.8429$$

- (b) Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.

Given: $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$.

Step 1: Find the Sum of Squared Errors (SSE)

$$R^2 = 1 - \frac{SSE}{SST}$$

$$\frac{SSE}{SST} = 1 - R^2$$

$$SSE = SST(1 - R^2)$$

Substitute the given values:

$$SSE = 725.94 \times (1 - 0.7911)$$

$$SSE = 725.94 \times 0.2089$$

$$SSE \approx 151.6489$$

Step 2: Calculate the estimated variance of the error term ($\hat{\sigma}^2$) The formula for $\hat{\sigma}^2$ in a simple linear regression (which has two parameters, β_1 and β_2) is:

$$\hat{\sigma}^2 = \frac{SSE}{N - 2}$$

Substitute the values:

$$\hat{\sigma}^2 = \frac{151.6489}{20 - 2}$$

$$\hat{\sigma}^2 = \frac{151.6489}{18}$$

$$\hat{\sigma}^2 \approx 8.4249$$

- (c) Suppose that a simple regression has quantities $\Sigma(y_i - \bar{y})^2 = 631.63$ and $\Sigma \hat{e}_i^2 = 182.85$, find R^2 .

Given: $\Sigma(y_i - \bar{y})^2 = 631.63$ and $\Sigma \hat{e}_i^2 = 182.85$.

Step 1: Identify the components

- $\Sigma(y_i - \bar{y})^2$ is the definition of SST , so $SST = 631.63$.
- $\Sigma \hat{e}_i^2$ is the definition of SSE , so $SSE = 182.85$.

Step 2: Calculate R^2

Using the formula relating R^2 to SSE and SST :

$$R^2 = 1 - \frac{SSE}{SST}$$

Substitute the values:

$$R^2 = 1 - \frac{182.85}{631.63}$$

$$R^2 = 1 - 0.2895$$

$$R^2 \approx 0.7105$$

Question 3

We have five observations on x and y . They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^5 \hat{e}_i^2 = 3.6$, $\sum_{i=1}^5 (x_i - \bar{x})^2 = 10$. Carry out this exercise with a hand calculator. Compute

- (a) the predicted value of y for $x_0 = 4$.

Preliminary Calculations

1. Sample Mean of x ($\bar{x}$):

$$\bar{x} = \frac{\sum x_i}{N} = \frac{3 + 2 + 1 - 1 + 0}{5} = \frac{5}{5} = 1$$

2. Estimated Error Variance ($\hat{\sigma}^2$):

We are given $N = 5$ and the sum of squared residuals $\sum \hat{e}_i^2 = 3.6$ (which is SSE). Degrees of freedom (df) = $N - 2 = 5 - 2 = 3$.

$$\hat{\sigma}^2 = \frac{SSE}{N - 2} = \frac{3.6}{3} = 1.2$$

3. Critical t-values (for $df = 3$):

- For a 95% interval ($\alpha = 0.05$, two-tailed): $t_{(0.025, 3)} = 3.1824$
- For a 99% interval ($\alpha = 0.01$, two-tailed): $t_{(0.005, 3)} = 5.8409$

Plug $x_0 = 4$ directly into the fitted least squares line:

$$\hat{y}_0 = 1.2 + 0.8x_0$$

$$\hat{y}_0 = 1.2 + 0.8(4)$$

$$\hat{y}_0 = 1.2 + 3.2 = 4.4$$

- (b) the $se(f)$ corresponding to part (a).

The formula for the variance of the forecast error is:

$$\widehat{var}(f) = \hat{\sigma}^2 \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right]$$

Substitute the known values ($x_0 = 4$, $\bar{x} = 1$, $\sum (x_i - \bar{x})^2 = 10$, $N = 5$, $\hat{\sigma}^2 = 1.2$):

$$\widehat{var}(f) = 1.2 \left[1 + \frac{1}{5} + \frac{(4 - 1)^2}{10} \right]$$

$$\widehat{var}(f) = 1.2 \left[1 + 0.2 + \frac{3^2}{10} \right]$$

$$\widehat{var}(f) = 1.2 [1 + 0.2 + 0.9]$$

$$\widehat{var}(f) = 1.2[2.1] = 2.52$$

The standard error of the forecast is the square root of the variance:

$$se(f) = \sqrt{\widehat{var}(f)} = \sqrt{2.52} \approx 1.5875$$

- (c) a 95% prediction interval for y given $x_0 = 4$.

The formula for a prediction interval is:

$$\hat{y}_0 \pm t_c \times se(f)$$

Using $t_{(0.025,3)} = 3.1824$:

$$4.4 \pm (3.1824 \times 1.5875)$$

$$4.4 \pm 5.05206$$

95% PI:

$$(-0.6521, 9.4521)$$

- (d) a 99 % prediction interval for y given $x_0 = 4$.

Use the exact same formula, but swap in the 99

$$4.4 \pm (5.8409 \times 1.5875)$$

$$4.4 \pm 9.272429$$

99% PI:

$$(-4.8724, 13.6724)$$

- (e) a 95 % prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

First, set our new x value to the mean: $x_0 = \bar{x} = 1$

Calculate the new predicted value:

$$\hat{y}_1 = 1.2 + 0.8(1) = 2.0$$

Calculate the new $se(f)$:

$$se(f) = \sqrt{1.2 \left[1 + \frac{1}{5} + \frac{(1-1)^2}{10} \right]}$$

$$se(f) = \sqrt{1.2 [1 + 0.2 + 0]}$$

$$se(f) = \sqrt{1.2 \times 1.2} = \sqrt{1.44} = 1.2$$

Calculate the new 95% PI:

$$2.0 \pm (3.1824 \times 1.2)$$

$$2.0 \pm 3.81888$$

$$(-1.8189, 5.8189)$$

Question 25

The file *collegetown* contains data on 500 houses sold in Baton Rouge, LA, during 2009–2013. Variable descriptions are in the file *collegetown.def*.

Table 1: Regression Results for Housing Prices in Baton Rouge (2009–2013)

Dependent Variable:	Model (a)	Model (b)	Model (c)
	Log-Linear $\ln(PRICE)$	Log-Log $\ln(PRICE)$	Linear $PRICE$
Intercept	4.3939*** (0.04329)	2.0497*** (0.1580)	-115.4236*** (13.0882)
$SQFT$	0.03604*** (0.00149)		13.4029*** (0.4492)
$\ln(SQFT)$		1.0248*** (0.04839)	
Observations	500	500	500
Standard R^2	0.5417	0.4738	0.6413
Generalized R^2	0.6622	0.6445	—
Residual Std. Error	0.3400	0.3643	102.8

Note: Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

- (a) Estimate the log-linear model $\ln(PRICE) = \beta_1 + \beta_2 SQFT + e$. Interpret the estimated model parameters. Calculate the slope and elasticity at the sample means, if necessary.

1. Interpreting the Model Parameters

- The Slope Coefficient ($\hat{\beta}_2 = 0.036044$): Because this is a log-linear model (the dependent variable is logged, but the independent variable is not), we interpret the coefficient as an approximate percentage change. To do this, we multiply by 100.
 - Interpretation: For every 1-unit increase in $SQFT$, the house price increases by approximately 3.6%, while holding all else constant.
- The Intercept ($\hat{\beta}_1 = 4.393866$): This represents the predicted log-price of a house with $SQFT = 0$.
 - Interpretation: This does not have a practical economic interpretation, as a house with zero square feet does not exist. It strictly acts as an anchor for the regression line.

2. Interpreting Slope and Elasticity at the Mean

Because the exact dollar impact of adding square footage changes depending on how big the house already is in a log-linear model, we evaluate the slope and elasticity at the sample averages:

- Slope at the Mean (slope_a = 9.02): The slope is the marginal effect, $\frac{d(PRICE)}{d(SQFT)}$. Taking the derivative yields $\beta_2 \times PRICE$. To calculate this at the sample mean, we use: $\hat{\beta}_2 \times \overline{PRICE}$. At the average house size and average house price for this sample, adding one more unit of $SQFT$ increases the expected price by 9.02 units (likely \$9,020, assuming the price variable is measured in thousands of dollars).
 - Elasticity at the Mean (elast_a = 0.983): Elasticity is $\frac{d(PRICE)}{d(SQFT)} \times \frac{SQFT}{PRICE}$. Substituting the slope gives $(\beta_2 \times PRICE) \times \frac{SQFT}{PRICE} = \beta_2 \times SQFT$. At the sample mean, this is $\hat{\beta}_2 \times \overline{SQFT}$. Elasticity measures a percentage-to-percentage relationship. At the sample mean, a 1% increase in a house's square footage is associated with a 0.983% increase in its price.
- (b) Estimate the log-log model $\ln(PRICE) = \alpha_1 + \alpha_2 \ln(SQFT) + e$. Interpret the estimated parameters. Calculate the slope and elasticity at the sample means, if necessary.

1. Interpreting the Model Parameters

- The Coefficient / Elasticity ($\hat{\alpha}_2 = 1.02483$): In a log-log model, the coefficient directly measures the elasticity, α_2 is constant across all values, so the elasticity at the sample mean is just $\hat{\alpha}_2$.
 - Interpretation: For every 1% increase in a house's square footage, the price increases by approximately 1.025%, holding all else constant. Because this value is very close to 1, we can say the relationship is roughly unit-elastic (a proportional increase in size leads to an almost identical proportional increase in price).
- The Intercept ($\hat{\alpha}_1 = 2.04965$): This represents the predicted $\ln(PRICE)$ when $\ln(SQFT) = 0$. Since $\ln(1) = 0$, this is the predicted log-price for a house with exactly 1 square foot.
 - Interpretation: Just like in part (a), this does not have a useful real-world economic interpretation and simply anchors the regression line.

2. Interpreting the Slope at the Mean

- Slope at the Mean (slope_b = 9.40): From the elasticity formula, Slope = Elasticity $\times \frac{PRICE}{SQFT} = \alpha_2 \times \frac{PRICE}{SQFT}$. At the sample mean, calculate: $\hat{\alpha}_2 \times \frac{\overline{PRICE}}{\overline{SQFT}}$. Even though the elasticity is constant everywhere in a log-log model, the slope (the absolute dollar change) depends on where you are on the curve. At the average house size and price for this dataset, adding one more square foot is associated with an expected price increase of 9.40 units (e.g., \$9,400). Notice how closely this aligns with the 9.02 unit slope from the log-linear model in part (a).
- (c) Compare the R^2 value from the linear model $PRICE = \delta_1 + \delta_2 SQFT + e$ to the “generalized” R^2 measure for the models in (b) and (c).

According to the Table 1, the Log-Linear model (a) provides the best overall fit for the data in terms of explaining the variation in actual house prices. It captures approximately 66.2% of the variation in *PRICE*, outperforming both the linear model (64.1%) and the log-log model (64.5%). This quantitative comparison provides strong preliminary evidence that transforming the dependent variable was a mathematically sound choice.

- (d) Construct histograms of the least squares residuals from each of the models in (a)–(c) and obtain the Jarque–Bera statistics. Based on your observations, do you consider the distributions of the residuals to be compatible with an assumption of normality?

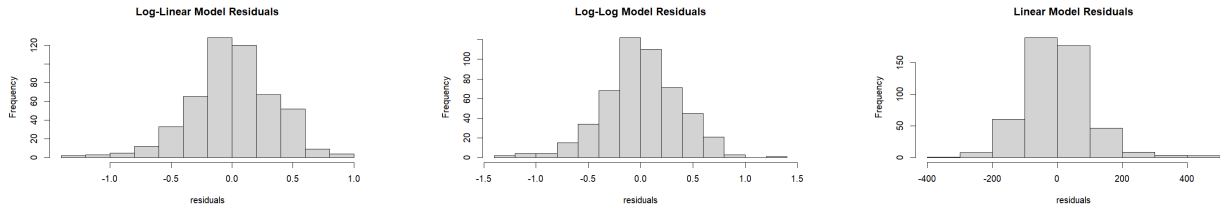


Figure 1: Histograms of the least squares residuals for different models

Table 2: Jarque-Bera Test Statistics for Residual Normality

Model	Functional Form	JB Statistic (χ^2)	df	p-value
(a)	Log-Linear	26.679	2	1.61×10^{-6}
(b)	Log-Log	14.279	2	0.000793
(c)	Linear	221.110	2	$< 2.2 \times 10^{-16}$

Note: The null hypothesis (H_0) is that the residuals are normally distributed.

The Jarque-Bera (JB) test evaluates whether a sample data set matches a normal distribution based on its skewness and kurtosis.

- Null Hypothesis (H_0): The residuals are normally distributed.
- Alternative Hypothesis (H_1): The residuals are not normally distributed.

Interpretation:

For all three models, the p-values from your JB tests are extremely small (far below standard significance levels like 0.05 or 0.01). Therefore, strictly speaking, we must reject the null hypothesis for all three models. Based on the formal test, the distributions of the residuals are not strictly compatible with the assumption of normality.

However, look at the JB statistic (X-squared) and the histograms themselves:

1. Linear Model: The JB statistic is massive (221.11), and the histogram shows a clear right skew (a long tail on the right side).
 2. Log-Linear and Log-Log Models: The JB statistics are drastically reduced (26.68 and 14.28, respectively). Visually, the histograms for the log-transformed models look much more symmetric and bell-shaped. While they still fail the strict statistical test (often due to slightly "fat tails" or excess kurtosis common in financial/pricing data), the log transformations have brought the distributions much closer to normality than the level-level linear model.
- (e) For each of the models in (a)–(c), plot the least squares residuals against $SQFT$. Do you observe any patterns?

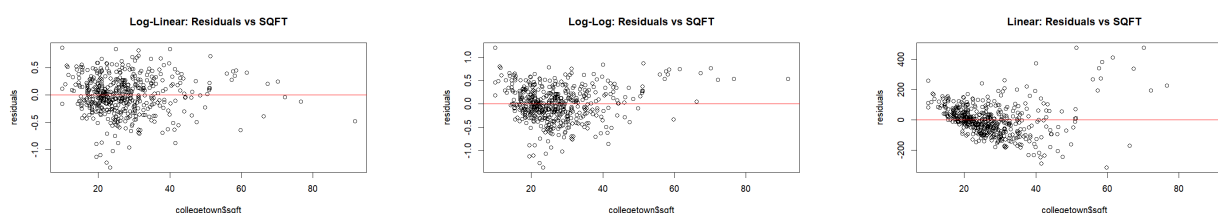


Figure 2: least squares residuals plot for different models

When plotting residuals against the independent variable ($SQFT$), you are looking for patterns. Ideally, the residuals should look like a random cloud of points evenly dispersed around the red zero line, indicating constant variance (homoskedasticity).

Interpretation:

1. Linear Model: There is a very clear pattern here. As $SQFT$ increases (moving to the right on the x-axis), the vertical spread of the residuals becomes much wider. This classic "funnel" or "cone" shape is a textbook example of heteroskedasticity. It tells us that our linear model's predictions are much less precise for large houses than for small houses.
 2. Log-Linear and Log-Log Models: Taking the natural log of $PRICE$ has effectively solved the heteroskedasticity problem. In both of these plots, the vertical spread of the residuals is relatively constant across the entire range of house sizes. The funnel shape has disappeared, and the variance has been successfully stabilized.
- (f) For each model in (a)–(c), predict the value of a house with 2700 square feet.

Table 3: Predictions and 95% Prediction Intervals for a 2,700 sq. ft. House ($SQFT = 27$)

Model	Functional Form	Predicted PRICE	95% Prediction Interval
(a)	Log-Linear	226.9817	[109.7685, 418.1167]
(b)	Log-Log	243.1514	[111.1406, 465.8406]
(c)	Linear	246.4557	[44.2773, 448.6341]

Note: PRICE values are estimated in thousands of dollars.

- (g) For each model in (a)–(c), construct a 95% prediction interval for the value of a house with 2700 square feet.

See Table 3.

- (h) Based on your work in this problem, discuss the choice of functional form. Which functional form would you use? Explain.

Discussing and Choosing the Functional Form

Conclusion: The **Log-Linear model (a)** is the superior functional form for this dataset and should be the model of choice. (The log-log model is a close second, but the linear model is definitely the worst).

1. Superior Goodness-of-Fit (Part c)

When comparing the models on a level playing field using the Generalized R^2 , the Log-Linear model explained the highest proportion of the variation in actual house prices (66.2%), outperforming both the linear model (64.1%) and the log-log model (64.5%).

2. Improved Normality of Errors (Part d)

While none of the models strictly passed the Jarque-Bera test for perfect normality, the log transformations massively reduced the JB test statistic (from a severe 221.11 in the linear model down to 26.68 and 14.28). The histograms demonstrate that transforming the dependent variable brings the error distribution much closer to the normal bell curve required for valid hypothesis testing.

3. Resolution of Heteroskedasticity (Part e)

The most critical failure of the standard linear model is the severe heteroskedasticity evident in its residual plot. The variance of the errors increases dramatically as $SQFT$ increases, which violates a core assumption of Ordinary Least Squares (OLS) and invalidates standard errors. Taking the natural log of $PRICE$ in the Log-Linear (and Log-Log) model successfully stabilized this variance, resulting in a homoskedastic residual plot.

4. More Realistic Forecasting (Parts f & g)

Because the linear model suffers from heteroskedasticity, its prediction interval for a large house ($SQFT = 27$) is excessively wide and unrealistic, predicting a lower bound of roughly \$44,2773. The Log-Linear model, benefiting from constant variance,

provides a much more constrained and economically sensible 95% prediction interval of approximately \$109,768 to \$418,117.